

Building Trust and Transparency in AI: A Review of Explainable AI and its Ethical Implications

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Abstract— As Artificial Intelligence (AI) becomes increasingly integrated into critical sectors, the need for transparency and trust in AI systems has grown significantly. This paper presents a systematic review of Explainable AI (XAI) and its role in aligning AI development with human values, particularly addressing the ethical concerns surrounding fairness, accountability and bias. XAI refers to a set of techniques and methods used to make the decision-making processes of AI systems more transparent and understandable to humans. XAI differs from conventional AI methods by empowering users to monitor and interpret AI outputs, fostering trust, and mitigating concerns about opaque decision-making processes. By focusing on the intersection of XAI and human-centered design, this review highlights the potential of XAI to enhance the ethical use of AI, contributing to the creation of transparent, responsible, and socially beneficial AI systems. The study also examines how XAI can support the ethical dimensions of sustainable development goals by driving the responsible development of AI technologies for sustainable infrastructures. Our findings emphasize that incorporating human values into AI design can promote organizational transparency, build public trust, and align AI behaviour with societal expectations. This review contributes to the broader literature on ethical AI, suggesting that future AI systems should perform effectively and act as trustworthy decision-makers that adhere to core societal principles.

Keywords— *Explainable Artificial Intelligence, Transparency, Biasness, Accountability, Sustainable Development, Trustworthiness*

I. INTRODUCTION

AI has recently undergone tremendous development, especially deep learning (DL) techniques, demonstrating remarkable performance in many areas. However, deep learning models' opacity or the 'black box' remains a critical challenge. Explainable Artificial Intelligence (XAI) can assist in reducing the gap between the theory of operation of AI and human comprehension [1]. XAI aims to explain the DL models, a branch of AI, to enable an understanding of its output. XAI techniques can be described as interpreters that translate the reasoning of a model into something understandable to humans. This not only builds confidence in the outcomes produced by AI but also offers explanations of how they arrived at such outcomes. XAI aids in uncovering the possible bias inherent in the training process and the algorithm used in the AI model. This bias can lead to prejudicial state action contrary to other virtues like fairness and equality for all human beings. For instance, an AI model applied to grant loans may discriminate against a particular group if trained on

prejudicial data. XAI may aid in locating such patterns in data, enabling the developer to maintain the data integrity and the model to adhere to the non-discrimination norms. Transparency is made in the sense of XAI by 'opening the lid' on AI's 'black box' [2]. Transparency can make XAI accountable for the actions and decisions of developing and using AI systems [3]. When the concept of XAI is applied to the functioning of AI, its decision-making process and compliance with human values can be understood, and an ethical basis can be built. This leads to the responsible use of these models within AI and the implementation of the safety protocols that protect these rights and the overall welfare of society.

Table I defines concepts such as understandability, accountability, explainability, and transparency used in the study. These terms sometimes have different connotations in different research studies and investigations. The Sustainable Development Goals (SDGs) consist of 17 significant objectives agreed upon by all United Nations member states in 2015, with a vision to achieve them by 2030. These goals include eradicating poverty and hunger, promoting social, economic, and political inclusiveness, and addressing climate change issues. Specifically, SDG 9 focuses on developing infrastructures for sustainable industrialization, production, and technological advancement. In this context, Explainable AI (XAI) is crucial in supporting these objectives by promoting transparency and fairness in AI systems, which fosters non-discrimination and sustainability [2]. XAI techniques can be applied to data to build AI models that suggest ways for factories to reduce energy consumption, ultimately contributing to a more sustainable future. Consequently, XAI can serve as a vital tool for ensuring the responsible use of technology in industries, aligning with the goals of SDG 9. Additionally, SDG 9 emphasizes the importance of technology in achieving sustainable development. XAI contributes to building trust and fairness by using AI models necessary for achieving these goals. By making AI models more understandable, XAI enhances public acceptance and ensures that the benefits of technological advancements are accessible to all [4]. Moreover, XAI facilitates the identification of biases within AI systems used for tasks such as resource allocation. By uncovering the origins of these biases, corrections can be made to ensure fairness and sustainability in outcomes [5].

XAI can further enhance the sustainability of AI by providing insights into how AI systems optimize resource allocation, energy usage, and waste minimization.

TABLE I: CONCEPT DEFINITIONS

Terms	Definition
Understandability	It refers to a model's features that allow people to understand how it works without looking "under the hood."
Explainability	It is the capacity to show the inner workings of AI models, making it clear to humans how these models reach their decisions.
Transparency	It uncovers the AI black box, allowing users to understand its reasoning, creating trust and ethical alignment.
Trustworthiness	It transcends accuracy. XAI fosters trust by ensuring transparency, fairness, and accountability, aligning AI with human values.
Accountability	It emphasizes the need to credit AI decisions to individuals. XAI, thus, enables us to assign accountability to AI system developers and users by explaining how models produce those outputs.
Fairness	It focuses on the problem of decreasing the impact of bias in AI models. Machine learning is used through XAI methods to detect prejudiced data or algorithm bias and produce fair and ethical results.

The review is guided by various key research questions, such as: What are the many objectives of XAI, and how do they contribute to building trustworthy and ethical AI? How does XAI promote transparency and accountability in AI decision-making processes? What are the different explainability techniques used in XAI, and how do they contribute to mitigating potential biases in AI models? How can XAI enhance collaboration between humans and AI systems while ensuring alignment with human values? How can we use XAI to make innovation more sustainable and contribute to Sustainable Development Goals?

This study reviews the current landscape of XAI, examining its applications, methodologies, and alignment with human values like transparency, fairness, and accountability. The research objectives of the study are:

- To review the existing literature on XAI
- Analyse the integration of XAI and human values
- Explore the role of XAI in achieving sustainable development goal 9
- Provide recommendations for future research and policy

A brief outline of the research paper includes sections from I to V. Section I introduces XAI, SDG and human values. Section II provides the literature review of XAI, its literature search strategy, technical perspective, and alignment with human values. Section III contains the frameworks of XAI. Section IV includes the results of the research on topics viz. XAI, Ethical AI, Interpretable AI, and Explainable AI. Section V is the conclusion of the paper and talks about future insights into XAI.

II. LITERATURE REVIEW

Researchers, policymakers, and practitioners have paid more attention to combining Artificial Intelligence (AI) and human values in recent years. As AI technologies become a larger part of different areas in society, we must ensure that these systems align with human values like openness, fairness, responsibility, and reliability. This

literature review talks about the theoretical background of XAI, its technical perspective, and aligning AI with human values.

2.1 XAI: Theoretical Background

Incredible developments are witnessed in AI study, especially in deep learning models that have transformed and developed many fields. However, a critical challenge persists. The interpretability of the models is always an issue, an infamous black box. In a black box model, only the input and the output are visible, but the internal process remains unknown. Its internal work is highly convoluted, thus preventing people from having faith in their decision-making, especially in areas like health and finance. This lack of transparency necessitates XAI [6] a discipline closing the gap between opaque AI systems and humans. Benefits of prompt transparency in XAI provide insight into the decision-making process of the AI system, which is vital in sectors such as healthcare and finance because clear decision-making fosters reliability and accountability [7]. The frameworks for XAI, i.e. Grad-CAM and SHAP, have been shown to enhance interpretability and feature-relevant selection processes among models [8]. XAI enhances trust and increases usability since decision tracking is possible—an essential aspect of ethical compliance. Moreover, XAI makes debugging and improvement easier, paves the way for making AI systems more or less impartial, and fulfills certain laws such as the GDPR (General data protection regulation).

While complex models may be more accurate, but they may be complex and difficult to explain. XAI methods can also increase computation time as explanations are created. In addition, these explanations can sometimes be beyond a normal person's understanding, thus proving almost useless. XAI is not always comprehensive, and inadequate explanations may not be sufficient in some cases. This subsection is further divided into transparency and causality, levels of transparency, and transferability.

Table II contains the three categories of transparency models.

Table II: TYPES OF AI MODELS

Model type	Reference No. and Year	Description
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Black box	[3], 2018	This is the least transparent model where decisions are made without any explanation the internal workings are too complex for humans to interpret easily.
White box	[11], 2019	It provides partial transparency by showing which features influenced decisions but does not fully explain the reasoning behind feature importance
Grey box	[12], 2016	The most transparent model with easily interpretable logic behind its predictions, it is highly interpretable but may lack accuracy for complex tasks.

These models based on the degree of explanation they provide and the nature of the AI.

2.1.1 Transparency and Causality- A fundamental goal of XAI is to break down the "black box" nature of AI and shed some light on the inner workings of AI models. However, XAI goes beyond the point of explaining outputs; it also digs into the "why" behind the "what" by providing explanations, XAI allows us to understand the factors most influencing a specific decision [9]. It is very important for building trust and ensuring that AI decisions are based on sound logic and not just correlations [10].

2.1.2 Levels of Transparency- The levels of transparency can be divided into several types. The major levels of transparency are simulatability, decision-making, and algorithmic transparency. The characteristics of simulatability are concerned with accurately predicting the model's behavior for pristine input scenarios. It lets one "try out" the models and understand how they would behave given a set of data inputs. Using the model with some unexpected values helps to determine how the model will resolve unexpected data or possible errors. Methods include feature importance analysis, in which we ranked features based on how much they affected the model's output. Decision rule extraction for rule-based models pulls out the rules that guide the AI system's decisions. Sensitive analysis examines how the model's results change when we tweak individual input features. This aspect digs into the algorithmic transparency of the model's specific algorithms and their deep-rooted biases. It involves breaking down the logic and structure of the algorithm in the model to help us spot potential bias sources built into its design. These include adding more diverse data, using algorithms that remove bias, and training models with fairness in mind.

2.1.3 Transferability- In the real world, we often need models that work well even when the data differs. XAI methods can help us determine if a model can be used in new situations. This helps developers understand how well the model might work in new settings. It also lets them make specific changes when using models in new places. For instance, an XAI technique reveals that a loan-approval model is heavily influenced by one of the income variables. If this factor isn't present in the new place where the model will be used, XAI can search for other factors that can maintain model relevancy.

2.2 XAI: Technical Perspective

Figure 1 shows the architecture of a XAI model. It includes all components, such as raw training data, which can be structured (tabular) or unstructured (text, images). The model represents the AI system being trained, with model

fitting referring to the process of learning from data and fine-tuning internal parameters to optimize task performance. A surrogate model, a simpler representation of the original, is created after training to provide a more interpretable understanding of the model's behavior.

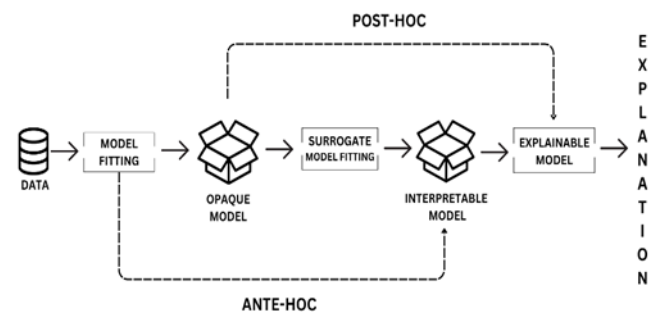


Figure 1. XAI Architecture Diagram

There are techniques for ante-hoc and post-hoc methods to realize the explainability of AI models. Ante-hoc techniques inherently have explainability integrated into them right from the design stage and use models such as decision trees or linear models, where decision-making is intelligible, and effects of features are easily comprehensible. On the other hand, post hoc analysis is employed after the training of the model to explain the internal computation of the model. Such decision aids include sensitivity analysis that determines the impacts of output given input variations, feature importance analysis to determine critical factor indicators of outcome, and counterfactual explanations that outline marginal changes in input that significantly alter outputs. These are more important when choosing between these techniques and it has been seen that post hoc methods are usually ideal when high interpretability is desired while ante-hoc methods are normally computationally intensive.

2.3 Aligning AI with Human Values

Ethical standards are the rules of behavior that a person believes are appropriate and correct and include norms, including honesty, integrity, responsibility, compassion, respect, and cooperation. The following discusses why there is a need to ensure that artificial intelligence is properly integrated with human values. XAI outlines how it can be connected to human ethics.

Ethical Alignment: Black box solutions increase the threat of preserving prejudice or ethical misconduct from a data set or algorithms utilized in the model [13]. XAI breaks down the decision-making process. This ensures that AI complements human values, including fairness, non-discrimination, and respect for human rights [14].

Accountability and Responsibility: XAI fosters accountability. This way, one can identify developers' and users' accountability for bias and undesired outcomes in AI decisions [15].

Risk Mitigation: Prejudice or unexpected blunders may occur when data or models are used to construct AI systems, which can negatively affect society. XAI also helps regulate the use of AI-based solutions [16]. By recognizing this, one can act to mitigate threats and guarantee that AI stays aligned with human values to avoid social harm.

Table III depicts human values and their alignment with XAI. It includes human values, such as honesty, integrity, responsibility, compassion, cooperation, and respect.

III. METHODOLOGY

This section systematically presents the major frameworks in Explainable AI. Table IV contains frameworks, a brief description, and citations from the literature supporting the

specific framework. This approach provides a straightforward description of the fundamental concepts that embrace XAI. Figure 2 shows all the techniques of explainable artificial intelligence (XAI) and how they are all connected to XAI.

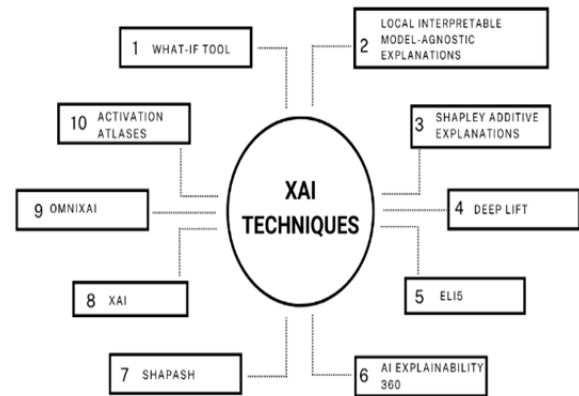


Figure 2. XAI technique

TABLE III: HUMAN VALUES AND THEIR ALIGNMENT WITH XAI

Human Values	Definition	Reference No.	Insight
Honesty	Truthfulness, integrity, and transparency in our actions and communication.	[3], [14], [17]	XAI has an impact on openness by showing how AI makes choices. This helps AI systems work honestly and truthfully.
Integrity	Upholding strong moral principles and acting with honesty and fairness.	[1], [18], [19]	XAI ensures AI systems follow ethical rules and moral ideas, which keeps AI trustworthy.
Responsibility	Taking ownership of our actions and the Consequences.	[2], [12], [15]	When AI gives clear reasons for its choices, XAI lets makers and users take charge of what AI systems do and decide.
Compassion	Empathy, understanding, and a desire to reduce suffering.	[7], [20], [21]	In health care, XAI ensures people can understand and trust AI choices. This leads to kind results that think about patients' well-being and help them feel better.
Cooperation	Working together for mutual benefit and achieving common goals.	[13], [22], [23]	XAI helps AI systems and people work together by giving reasons people can use and make sense of. This builds trust and teamwork between humans and machines.
Respect	Treating others with dignity, consideration, and appreciation for their inherent worth.	[16], [24], [25]	XAI respects users and stakeholders by ensuring AI decisions are clear and defensible. This honors the right to grasp how the system reaches conclusions.

TABLE IV: XAI FRAMEWORKS

Framework/Tool	Definition	Reference No.
What-If Tool	An open-source tool developed by Google that helps users analyze machine learning systems with minimal coding, assess performance, and visualize models.	[26]
Local Interpretable Model-Agnostic Explanations (LIME)	A technique that explains predictions by learning an interpretable model localized around the prediction to clarify any classifier's decisions.	[12]
Shapely Additive Explanations (SHAP)	A framework that assigns importance values to each feature in a prediction, offering a new class of additive feature importance measures.	[27]
Framework/Tool	Definition	Reference No.
Deep LIFT	A method to dissect neural networks by tracing the influence of each input feature on the output, using reference activations to assign contribution scores.	[28]
ELI5	A Python tool that explains and debugs machine learning classifier predictions, compatible with several frameworks like LightGBM and XGBoost.	[29]
Shapash	A Python module that makes machine learning models accessible and interpretable by providing visualizations and summary explanations.	[27]
AI Explainability 360 (AIX 360)	An open-source toolkit that includes a wide range of algorithms for explainability in machine learning and data models.	[22]
XAI (ML library)	A machine learning library focused on AI explainability, helping users analyze and evaluate models across phases like data analysis and model evaluation.	[14]

Omni XAI	An open-source XAI library developed by Salesforce was designed to make model explainability accessible with an easy-to-use interface for various data and models.	[23]
Activation Atlases	Builds on feature visualization by compiling and displaying representations in neural networks' hidden layers, showing how neurons interact collectively.	[30]

IV. RESULTS

The data gathered from "app.dimensions.ai" includes the number of papers published from Jan 1, 2015 to July 31, 2024, on topics such as 'Ethical AI', 'Interpretable AI,' 'Responsible AI,' and 'Explainable AI.'

Figure 3 shows the trends of four AI topics from 2015 to 2024, and it demonstrates the growth of the subjects over the period. The number of entries under ethical AI has grown substantially from 32 to 4252 between 2015 and 2023, showing awareness level and focus has grown; mid-2024 already represented 5214 indicating a greater year-on-year growth than in 2023. Interpretable AI began with only 7 entries in 2015 but increased year on year up to 823 in 2023 reducing again to 413 in 2024, possibly undercounted as the total reached 745 by the mid-year. Accountable AI records increased slightly from 27 to 1633 from 2015 to 2023; mid-2024 entries elevated in 1829 and imply continued public concern for responsible AI despite slower communiqué submission. Explainable AI activity experienced a splendid rise from nine submissions in 2015 to 2627 in 2023 due to a spurt in the number of studies between 2017 and 2023. As for mid-year data for 2024, even though the reporting is partial, it is impossible to exclude the fact that attention will remain focused on this area.

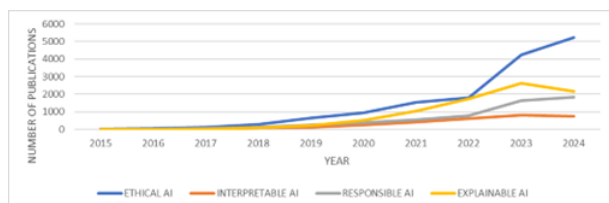


Figure 3. Comparison of Ethical, Interpretable, Responsible, and Explainable AI Papers Publishing Trends

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V. CONCLUSION

This systematic review proves how XAI assists in ensuring that AI technologies are aligned with human values and improve transparency. The increase in discussions about XAI demonstrates that people begin to recognize that AI systems have to be explainable and accountable. In this way, new technological ideas will be beneficial to society. The massive rise in discussion sessions on XAI suggests a collective effort toward tackling the role and problems of transparency and interpretability. It aligns with SDG 9, which aims to enhance the development of inclusive and sustainable infrastructure. Introducing human values into artificial intelligence is the foundation for attaining good and sustainable outcomes. Thus, XAI plays a crucial function in ensuring that AI technology or system is favoured and developed in a way that is understandable and aligned with society's standards. Despite progress, these problems have not been fully addressed. These include, for instance, searching for a balance between the model's accuracy and its ability to be described and implemented. It is through handling such problems that one can extend research and mobilize new arguments. Therefore, future work must be directed toward creating strong guidelines. Having well-defined guidelines for applying XAI increases research openness and aligns AI technologies with ethical regulations and sustainable development goal number 9.

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